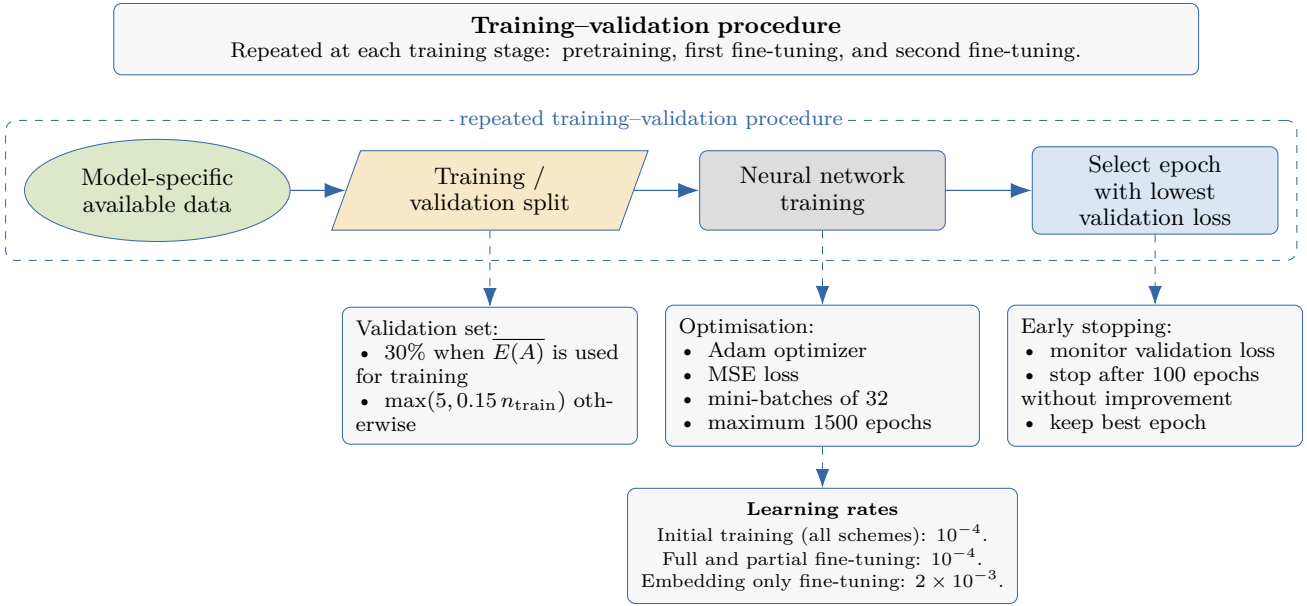

Training–validation procedure
Repeated at each training stage: pretraining, first fine-tuning, and second fine-tuning.
repeated training–validation procedure
Model-specific available data
Training / validation split
Neural network training
Select epoch with lowest validation loss
Validation set:
• 30% when $\overline{E(A)}$ is used for training
• $\max(5, 0.15\, n_{\text{train}})$ otherwise
Optimisation:
• Adam optimizer
• MSE loss
• mini-batches of 32
• maximum 1500 epochs
Early stopping:
• monitor validation loss
• stop after 100 epochs without improvement
• keep best epoch
Learning rates
Initial training (all schemes): 10^{-4}.
Full and partial fine-tuning: 10^{-4}.
Embedding only fine-tuning: 2×10^{-3}.